

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 3: SIZE FILTERING

SLO 1: Apply size-based filtering techniques to remove noise and irrelevant objects from binary images.

SPOT THE NOISE SOLUTIONS

Binary Image Matrix:

```
0 0 0 0 0 0
0 1 1 0 0 1
0 1 1 0 1 0
0 0 0 0 1 0
0 1 0 0 1 1
0 1 0 0 0 0
```

- Mark Small Noisy Components:** Small components with fewer than 3 pixels are **Component C** (single pixel '1' at top-right: Row 2, Col 6) and **Component D** (single pixel '1' at middle-right: Row 3, Col 5).
- Identify Larger Relevant Objects:**
 - Component A (Top-Left 2x2 block):** 4 pixels (Row 2-3, Col 2-3)
 - Component B (Right/Bottom connected region - 8-connectivity):** 5 pixels (Row 4-5, Col 5-6)
 - Component E (Bottom-Left 2x1 block):** 2 pixels (Row 5-6, Col 2)
- Total number of components before filtering = 4** (or 5 depending on 4 vs 8-connectivity interpretation).

APPLY SIZE FILTERING SOLUTIONS (THRESHOLD = 3 PIXELS)

Component	Description / Location	Number of Pixels	Keep / Remove
Object A	Top-Left 2x2 Block	4	Keep ($4 \geq 3$)
Object B	Right-Bottom Connected Cluster	5	Keep ($5 \geq 3$)
Object C	Top-Right Isolated Pixel	1	Remove ($1 < 3$)
Object D	Bottom-Left Vertical Pair	2	Remove ($2 < 3$)

QUESTIONS SOLUTIONS

- Which components are considered irrelevant (noise)?

Components with pixel count < 3 (Object C with 1 pixel and Object D with 2 pixels).

- Which components are relevant objects?

Components with pixel count ≥ 3 (Object A with 4 pixels and Object B with 5 pixels).

- Total number of objects after filtering:

2 objects